

SHREENIVAS A

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PROFESSIONAL SUMMARY

Results-driven Electronics & Communication Engineering student with hands-on industry experience as a PCB Designer and Embedded Systems Engineer. Proficient in full-cycle PCB design using Altium Designer — from schematic capture to routing, DFM, and validation. Skilled in embedded firmware development on STM32, ESP32, and Arduino platforms with expertise in CAN, SPI, and I2C communication protocols. Passionate about building robust, production-ready hardware-software solutions for industrial and IoT applications.

EDUCATION

B.E. Electronics and Communication Engineering (Final Year)
Chennai Institute of Technology, Chennai **CGPA: 8.19 / 10**

CORE COMPETENCIES

PCB Design & Layout	Embedded Systems	Communication Protocols	Programming	Sensor & Industrial IoT
Altium Designer, Proteus, LTSpice, Keil uVision	STM32, ESP32, Arduino, Raspberry Pi, Jetson Orin	CAN Bus, SPI, I2C, UART	Embedded C, Python (Core), BareMetal	Sensor Integration, Industrial Automation

PROFESSIONAL EXPERIENCE

- PCB Designer & Embedded Systems Engineer — ADHVX Industries** Nov 2024 – Jan 2025
- Designed a 2-layer DAC PCB containing 50+ components using Altium Designer for industrial sensors and a System on Module (SOM), applying multi-layer PCB layout and routing techniques in Altium Designer.
 - Performed embedded systems design and optimisation for sensor integration in industrial automation environments.
 - Applied PCB design best practices including DFM considerations, component placement, and basic signal integrity analysis to improve manufacturability and reduce routing issues.
- Engineering Trainee — Woory Automotive** May 2024 – Jun 2024
- Gained practical exposure to SMT processes, PCB segmentation, and precision soldering in a high-volume automotive production environment.
 - Optimised SMT loading accuracy to prevent component misalignment errors, improving yield rates.
- Research Author — Queueless Billing System (Academic Research)** May 2025 – Jun 2025
- Designed a smart billing architecture using embedded systems and IoT, implementing automated item detection with real-time bill generation and payment workflow.

KEY PROJECTS

- Rescue Wings | Disaster Management Drone System (MSME Project & Startup)**
- Led hardware integration of a dual-drone disaster management system — Collector Drone (camera + multi-sensor victim detection) and Delivery Drone (autonomous supply delivery based on real-time coordination).
 - Developed embedded firmware and communication interfaces for coordination and sensor data exchange between drones.
 - Developed under Rescue Wings, an MSME-registered startup initiative focused on drone-based disaster response solutions.
- Water Contamination Detection System | Embedded**
- Developed an IoT-based water quality monitoring system for real-time detection of contaminants, salt levels, & potential harmful substances in water
 - Integrated sensors, embedded controllers, and monitoring modules for continuous water quality analysis and alert generation.
 - Designed an automatic safety mechanism to shut off the contaminated water supply and display warning alerts for preventive action.
- Smart Laundry Management System | IoT + RFID**
- Built an end-to-end RFID+IoT tracking system to monitor laundry through the wash, dry, and delivery stages.
 - Integrated weight-based billing, automated SMS notifications, and a real-time web dashboard with QR-based payment.
- Anti-Drone Detection & Jamming System | RF + Embedded**
- Developed a real-time unauthorised drone detection system with RF-based jamming integration to neutralise threats, combining hardware signal processing with embedded control logic.

ACHIEVEMENTS & RECOGNITION

- Winner — Board Designer Competition, Hackster.io ([DAC Module for Industrial Sensors](#))
- Founder — Rescue Wings Startup (MSME Registered, Drone-based Disaster Management)
- Runner-up — Electrothon 2024 & SDG'25, Chennai Institute of Technology
- Shortlisted — Smart India Hackathon (SIH), IEEE Hackathon, FINTECH-A-THON at NIT, HumanAlze Hackathon
- Finalist — Ideathon 2024, Chennai Institute of Technology
- Participant — Hack4Purpose, Technica VIT

CERTIFICATIONS

- Cybersecurity Essentials - Cisco
- Energy Literacy Training
- Introduction to Industry 4.0 - NPTEL
- Introduction to Cybersecurity - Cisco
- Networking Essentials - Cisco
- Microsensors & Nanosensors - NPTEL